

Classifying Air Quality Index Levels Based on Pollutant Concentrations in Philippine Urban Environments Using Support Vector Machine, LightGBM, CatBoost, and Multilayer Perceptron Neural Network

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Abstract

Air pollution remains a significant environmental and public health concern in Philippine urban environments, where pollutant exposure is influenced by transportation activity, fuel combustion, industrial emissions, and dense urban development. This study presents a comparative machine learning approach for classifying Air Quality Index (AQI) severity levels using the PH Philippine Cities Air Quality Index Data 2025 dataset. The dataset contains city-level air quality records with pollutant concentration features, including CO, NO, NO₂, O₃, SO₂, PM2.5, PM10, and NH₃, along with temporal and city-based variables. The target variable, `main.aqi`, is treated as an ordinal multiclass label representing AQI severity levels on a 1-to-5 scale.

Four machine learning models were implemented and compared: Support Vector Machine, LightGBM Classifier, CatBoost Classifier, and Multilayer Perceptron Neural Network. Model performance was evaluated using accuracy, macro precision, macro recall, macro F1-score, weighted F1-score, and confusion matrix analysis. Results showed that LightGBM achieved the best overall performance, obtaining an accuracy of 0.9969 and a macro F1-score of 0.9704. CatBoost ranked second, while the MLP-NN and SVM baseline showed lower

performance. Feature importance analysis further indicated that pollutant concentration variables were more influential than temporal and city-based features. These findings suggest that gradient boosting models, particularly LightGBM, are effective for AQI severity classification using structured Philippine urban air quality data.

Keywords:

Air Quality Index, AQI Classification, Machine Learning, LightGBM, CatBoost, Support Vector Machine, Multilayer Perceptron Neural Network, Philippine Urban Air Quality

Introduction

Air pollution remains a significant environmental and public health concern in Philippine urban environments. It is associated with transportation activity, fuel combustion, industrial emissions, and dense urban development, all of which contribute to pollutant exposure in highly populated areas (Koons, 2024). Because air quality conditions affect public health, mobility, and urban planning, there is a continued need to strengthen how available air quality data are monitored, interpreted, and analyzed.

Air Quality Index (AQI) systems help communicate air quality conditions by translating pollutant measurements into severity levels that are easier for the public and decision-makers to understand. In the Philippines, several initiatives demonstrate the growing role of air quality monitoring in urban management. Quezon City, for example, expanded its monitoring capability by adding more air quality sensors and using the collected data to support policy interventions and public AQI reporting (Jcteamjoy, 2024). Similarly, EMB Region XI describes the Davao City Air Quality Monitoring Network as an urban-scale monitoring system intended to support air quality trending for researchers, decision-makers, and lawmakers (*Air Quality Management | EMB-XI*, n.d.). In Baguio City, Project MINERVA further illustrates the use of sensor-based and data-driven systems for smart-city monitoring, including air quality, water quality, mobility, and tourism-related indicators (Department of Science and Technology, n.d.).

Although these efforts show the increasing availability of air quality data, monitoring alone does not fully explain how pollutant concentration patterns correspond to AQI severity levels. Machine learning can support the analytical use of these data by modeling relationships between pollutant indicators and air quality categories. However, different algorithms rely on different assumptions and learning mechanisms, making comparative evaluation important.

This study classifies AQI levels in Philippine urban environments using the PH Philippine Cities Air Quality Index Data 2025 dataset. The dataset includes pollutant indicators such as PM_{2.5}, PM₁₀, CO, NO, NO₂, SO₂, O₃, and NH₃, along with temporal information and city-based variables. The

target variable, main.aqi, is treated as an ordinal multiclass label representing AQI severity levels. Four models are compared: Support Vector Machine, LightGBM Classifier, CatBoost Classifier, and Multilayer Perceptron Neural Network. Model performance is evaluated using precision, recall, F1-score, macro F1-score, confusion matrix analysis, and accuracy as a supplementary overall metric. The study also examines feature importance and classification behavior to analyze how different models respond to pollutant concentration patterns. Through this approach, the study aims to support environmental health monitoring by providing a comparative machine learning analysis of AQI severity classification in the Philippine context.

Related Work

A. Traditional Statistic Models for Air Quality Prediction

Air quality prediction remains a crucial field of research in machine learning due to its impact on public health, climate change, and environmental sustainability. In the pursuit of identifying relationships between air pollution and meteorological elements and quantifying the improvements from artificial intelligence (AI) implementation, Peinado et al. (2025) identified multiple linear regression (MLR) as the standard among 412 peer-reviewed articles. Through its ability to perform meteorological normalization, which reduces the influence of weather elements to measure increases in factors such as emissions, MLR stands as the “industry standard,” with a use case of 88.6%. This metric is supported by Gulati et al. (2023), demonstrating that MLR was capable of

detecting harmful particulate matter with an R^2 of 0.72.

Although multiple linear regression is widely used because of its interpretability and low computational cost, it remains limited in capturing dynamic interactions, multivariate dependencies, and non-stationary effects common in urban atmospheric environments (Peinado et al., 2025).

B. Machine Learning for Air Quality Prediction

Machine learning has been widely applied to air quality prediction because pollutant concentrations often exhibit nonlinear relationships influenced by environmental, temporal, and location-based factors. Liu et al. developed an AQI and pollutant concentration prediction approach using support vector regression and reported strong predictive performance, showing that kernel-based models can effectively capture nonlinear relationships in air quality data (Liu et al., 2019). Similarly, Leong et al. applied support vector machine methods to air pollution index prediction and emphasized the importance of parameter selection in improving model performance (Leong et al., 2020). These studies support the use of SVM-based models as classical machine learning approaches for air quality modeling.

Recent studies have also explored comparative machine learning approaches for AQI prediction. Tırnık presented a long-term AQI forecasting study using pollutant and meteorological variables and compared models such as XGBoost, LightGBM, and SVM, showing the usefulness of ensemble and kernel-based models for air quality prediction tasks

(Tırnık, 2025). These studies demonstrate that air quality modeling is often improved when multiple algorithms are compared under the same dataset and evaluation setting.

C. Gradient Boosting and Neural Models for Air Quality Modeling

Tree-based ensemble methods have become common in air quality studies because they can model nonlinear feature interactions and handle structured tabular data effectively. LightGBM is a gradient boosting decision tree framework designed for efficient training and high predictive accuracy on large datasets (Ke et al., 2017). CatBoost has also been applied in air quality prediction studies and has been reported to perform competitively among machine learning models for AQI prediction (Ravindiran et al., 2023). These models are relevant for air quality datasets that include heterogeneous variables such as pollutant readings, temporal attributes, and location-based features.

Neural network models have likewise been used for air quality prediction because of their ability to approximate nonlinear relationships among input variables. However, neural models may require careful preprocessing, feature scaling, and training control to avoid overfitting, especially when compared with tree-based boosting methods on tabular data. For this reason, comparing neural models with classical and ensemble models can provide useful insight into whether additional model complexity improves AQI classification performance.

D. Research Gap

Existing studies show that machine learning models can be effective for air quality prediction and classification.

However, many studies focus on continuous AQI regression, pollutant forecasting, or datasets outside the Philippine context. Fewer works emphasize multiclass AQI severity classification using structured Philippine urban air quality data while comparing classical, gradient boosting, and neural network models under a consistent evaluation pipeline.

To address this gap, the present study focuses on classifying the 1-to-5 main.aqi severity levels from the PH Philippine Cities Air Quality Index Data 2025 dataset. The study compares Support Vector Machine, LightGBM, CatBoost, and Multilayer Perceptron Neural Network using the same preprocessing, splitting, and evaluation framework. It also emphasizes macro F1-score and confusion matrix analysis to account for class imbalance and the ordinal nature of AQI severity levels, while using feature importance analysis to support environmental interpretability.

Model & Algorithm

This study formulates AQI level prediction as a supervised multiclass classification problem. Each record consists of pollutant concentration variables, temporal attributes, and location-based features, while the target variable is main.aqi. Since main.aqi is represented as a categorical AQI severity level, the models are trained to classify each observation into one of the AQI levels rather than predict a continuous AQI score.

A. Support Vector Machine

Support Vector Machine is used as a classical margin-based classifier. It assumes that classes can be separated by decision boundaries in a transformed feature space. Through kernel functions, SVM can model nonlinear boundaries between AQI severity levels, making it suitable for pollutant data

where class separation may not be strictly linear.

B. LightGBM

LightGBM is used as an efficient gradient boosting classifier. It builds decision trees sequentially, where each new tree attempts to correct the errors of the previous ensemble. This mechanism allows LightGBM to capture nonlinear relationships and feature interactions among pollutant concentration variables.

C. CatBoost

CatBoost is used as another gradient boosting classifier, with emphasis on its ability to handle categorical variables effectively. Since the dataset includes city-level information, CatBoost is useful for modeling location-based variation while reducing issues commonly associated with categorical encoding.

D. MLP

The Multilayer Perceptron Neural Network is used as a neural classification model. It consists of fully connected layers that learn nonlinear transformations of the input features. Hidden layers use nonlinear activation functions, while the output layer uses softmax activation to produce class probabilities for each AQI severity level. The final predicted class is selected based on the highest output probability.

Implementation

A. Dataset

This study uses the PH Philippine Cities Air Quality Index Data 2025 dataset from Kaggle. The dataset is organized into monthly files covering Philippine city-level air quality records for 2025. Since the

dataset is separated by month, the monthly CombinedData files were merged into a single master dataset before preprocessing, model training, and evaluation.

Each record represents an air quality observation for a specific city and timestamp. The dataset contains pollutant concentration variables, temporal information, city information, and the target AQI severity label. The target variable, main.aqi, is measured on a 1-to-5 ordinal scale based on the OpenWeather API standard utilized by the Kaggle dataset. Therefore, the study is formulated as a supervised multiclass classification task predicting a 1-to-5 severity level, rather than a continuous AQI regression problem.

The predictor variables consist of pollutant concentration features such as carbon monoxide, nitrogen monoxide, nitrogen dioxide, ozone, sulfur dioxide, fine particulate matter, coarse particulate matter, and ammonia. The dataset also includes datetime, which may be used to derive basic temporal features, and city_name, which represents the Philippine city associated with each air quality record.

Variables used in the study

Variable	Description
datetime	Timestamp of the air quality record. Used to derive temporal features such as month, day of week, and hour.
main.aqi	Target variable representing the AQI severity class. Treated as an ordinal multiclass

	label for classification.
components.co	Carbon monoxide concentration.
components.no	Nitrogen monoxide concentration.
components.no2	Nitrogen dioxide concentration.
components.o3	Ozone concentration.
components.so2	Sulfur dioxide concentration.
components.pm2_5	Fine particulate matter concentration, or PM2.5.
components.pm10	Inhalable particulate matter concentration, or PM10.
components.nh3	Ammonia concentration
city_name	Name of the Philippine city associated with the air quality record. Treated as a categorical city-based feature.

B. Data Preprocessing

After merging the monthly CombinedData files into a single master dataset, duplicate records were removed and missing values were inspected. Records with missing main.aqi values were excluded, while missing pollutant values were handled

through removal or imputation depending on their extent.

Column names may be standardized for easier processing, such as converting components.pm2_5 to pm2_5 and components.no2 to no2. The datetime column was converted into a proper datetime format, and simple temporal features such as month, day of week, and hour may be extracted. The raw datetime column will then be excluded from the final feature matrix.

The target variable, main.aqi, was separated from the predictor variables and excluded from the input features to prevent data leakage. Since main.aqi is already encoded as numeric AQI severity classes, no additional label encoding is required for the target. The city_name column was encoded as a categorical predictor, using one-hot encoding for models that require numerical input and native categorical handling for CatBoost where applicable.

Numerical feature scaling was applied for models sensitive to feature magnitude, particularly Support Vector Machine and Multilayer Perceptron Neural Network. Finally, the dataset were divided into stratified training, validation, and testing sets to preserve the distribution of AQI severity classes.

C. Train-Validation-Test Split

Since the dataset is organized into monthly CSV files for 2025, all monthly CombinedData files will first be merged into a single master dataset. After preprocessing, the combined dataset were divided into

training, validation, and testing sets using a stratified 70/15/15 split. Stratification was based on the target variable, main.aqi, to preserve the distribution of AQI severity classes across all partitions.

The training set was used to fit the models, the validation set was used for hyperparameter tuning and early stopping, and the testing set was reserved for final performance evaluation. The same data partitions were used across all models to ensure a fair comparison.

Since the study focuses on AQI level classification rather than future-month forecasting, a time-based split and sliding-window sequence generation will not be applied. Each row was treated as an independent air quality observation.

D. Model Training and Hyperparameter Tuning

The study will implement four classification models: Support Vector Machine, LightGBM Classifier, CatBoost Classifier, and Multilayer Perceptron Neural Network. Support Vector Machine will serve as the baseline model, while LightGBM and CatBoost will represent gradient boosting approaches for structured tabular data. The Multilayer Perceptron Neural Network will represent a neural network-based classifier.

A limited hyperparameter tuning process was conducted using the validation set to improve model performance while keeping computational requirements manageable. For SVM, tuning may include the kernel type, regularization parameter C, and gamma. For LightGBM and CatBoost,

tuning may include learning rate, tree depth, number of estimators, and number of iterations. For the MLP-NN, tuning may include hidden layer size, learning rate, batch size, and number of epochs.

The MLP-NN will use ReLU activation in the hidden layers and a softmax output layer for multiclass AQI classification. It was trained using the Adam optimizer and sparse categorical cross-entropy loss. Early stopping may be applied based on validation loss to reduce overfitting.

E. Evaluation Metrics

Model performance were evaluated using classification metrics appropriate for multiclass AQI level classification. The main metrics include precision, recall, F1-score, macro F1-score, and confusion matrix analysis. Accuracy will also be reported as a supplementary overall metric. Since main.aqi is ordinal and may be imbalanced, macro F1-score and class-level confusion matrix analysis were emphasized.

Evaluation Metrics Used in the Study

Metric	Purpose in the study
Accuracy	Measures the overall proportion of correctly classified AQI records. It were reported as a supplementary metric because it may be misleading if AQI classes are

	imbalanced.
Precision	Measures how reliable the model's predicted AQI classes are by identifying how many predicted instances of a class are actually correct.
Recall	Measures how well the model identifies actual AQI severity classes, especially less frequent or higher-severity AQI levels.
F1-score	Provides a balance between precision and recall for each AQI class.
Macro F1-score	Computes the average F1-score across all AQI classes, giving equal importance to each class regardless of class frequency. This is emphasized because AQI levels may be imbalanced.
Confusion Matrix	Shows class-level prediction errors and identifies which AQI severity levels are commonly misclassified. Since main.aqi is ordinal, this will also help determine whether errors occur between

	neighboring or distant AQI levels.
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F. Class Imbalance, Ordinality, and Interpretability

A key consideration in this study is the imbalance of AQI severity classes. In the dataset, lower AQI severity levels occur more frequently than higher-severity classes. Because of this, accuracy alone may not fully represent model performance, since a classifier may perform well on dominant classes while performing poorly on less frequent AQI levels. For this reason, macro F1-score and confusion matrix analysis were emphasized alongside overall accuracy.

Another consideration is the ordinal nature of the main.aqi label. Although the task is formulated as multiclass classification, the AQI classes follow a

meaningful severity order from lower to higher air quality risk. Therefore, model errors were interpreted not only as correct or incorrect predictions, but also based on whether misclassifications occurred between neighboring or distant AQI severity levels.

Interpretability was also considered because performance metrics alone do not explain which variables contributed most to classification decisions. To address this, feature importance from the best-performing model was examined. This allowed the study to identify whether pollutant concentration variables, temporal features, or city-based features contributed most to AQI severity classification.

Results

This study compared the performance of Support Vector Machine, LightGBM Classifier, CatBoost Classifier, and Multilayer Perceptron Neural Network in classifying AQI severity levels using pollutant concentration, temporal, and city-based features. Before model evaluation, the distribution of AQI severity classes was examined to determine whether the dataset was balanced. As shown in Fig. 1, the dataset is dominated by lower AQI classes, particularly class 1 and class 2, while higher-severity classes occur less frequently. This imbalance justifies the use of macro-averaged metrics, especially Macro F1-score, because accuracy alone may overstate model performance when majority classes dominate the dataset.

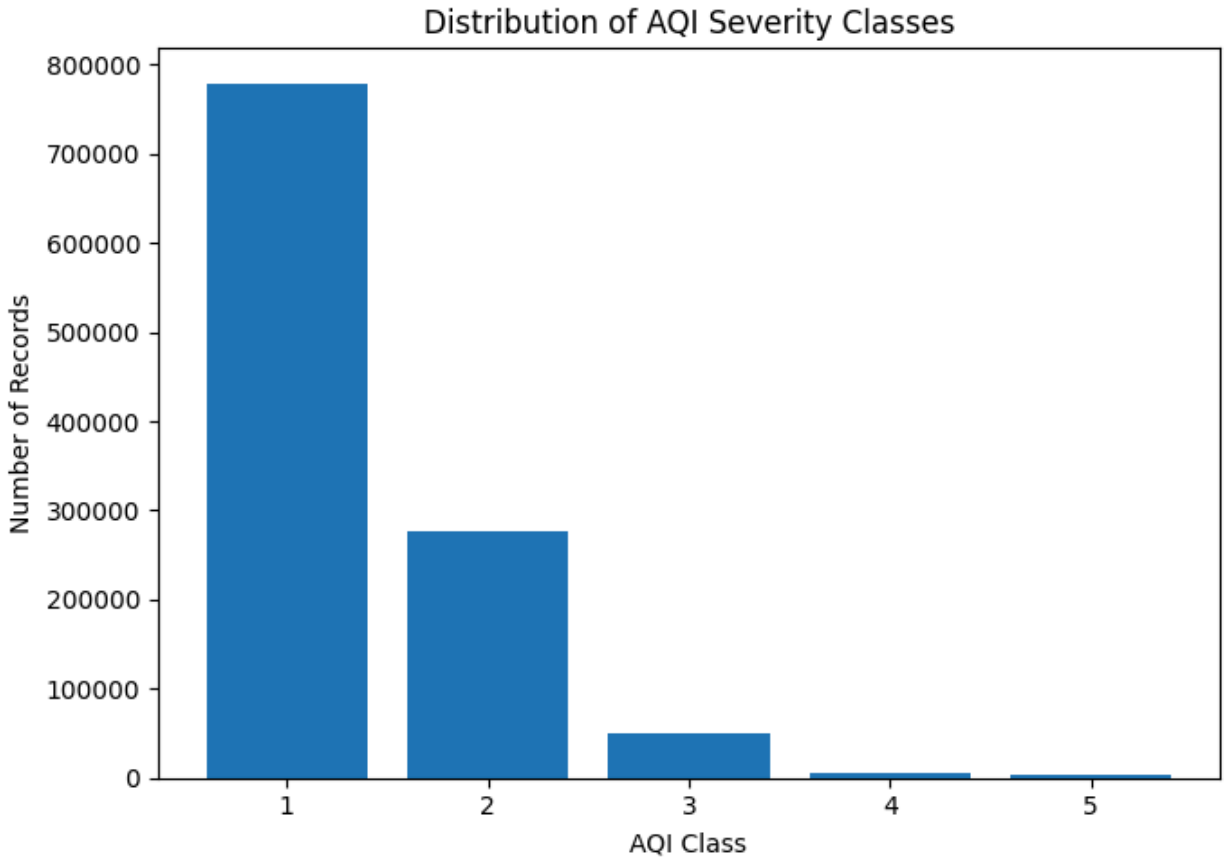


Fig. 1. Distribution of AQI severity classes in the cleaned dataset.

Table 1 presents the final test performance of the four classification models. Among all evaluated models, the LightGBM Classifier achieved the highest overall performance, with an accuracy of **0.9969**, macro precision of **0.9599**, macro recall of **0.9817**, macro F1-score of **0.9704**, and weighted F1-score of **0.9970**. These results indicate that LightGBM was the most effective model for classifying AQI severity levels from the selected pollutant concentration features.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
LightGBM	0.9969	0.9599	0.9817	0.9704	0.9970
CatBoost	0.9926	0.9118	0.9591	0.9338	0.9927
MLP-NN	0.9493	0.8926	0.8013	0.8381	0.9489

SVM	0.8461	0.6853	0.5687	0.6010	0.8431
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Table 1. Final Test Performance of AQI Classification Models

The CatBoost Classifier achieved the second-best result, with an accuracy of 0.9926 and macro F1-score of 0.9338. This shows that gradient boosting models performed strongly on the structured AQI dataset. However, LightGBM still outperformed CatBoost across all major macro-averaged metrics. The MLP Neural Network achieved an accuracy of 0.9493 and macro F1-score of 0.8381, indicating that it learned useful nonlinear patterns but did not match the performance of the boosting models. The Support Vector Machine served as the baseline model and obtained the lowest performance, with an accuracy of 0.8461 and macro F1-score of 0.6010. This suggests that the SVM baseline was less capable of capturing the nonlinear and imbalanced class structure of the AQI dataset.

The comparison of models based on Macro F1-score is shown in Fig. 2. This figure highlights the performance gap between the gradient boosting models and the baseline SVM. LightGBM achieved the highest Macro F1-score, followed by CatBoost, MLP-NN, and SVM. Since Macro F1-score gives equal weight to all classes, the result suggests that LightGBM was not only accurate on majority AQI classes but also performed well across less frequent severity levels.

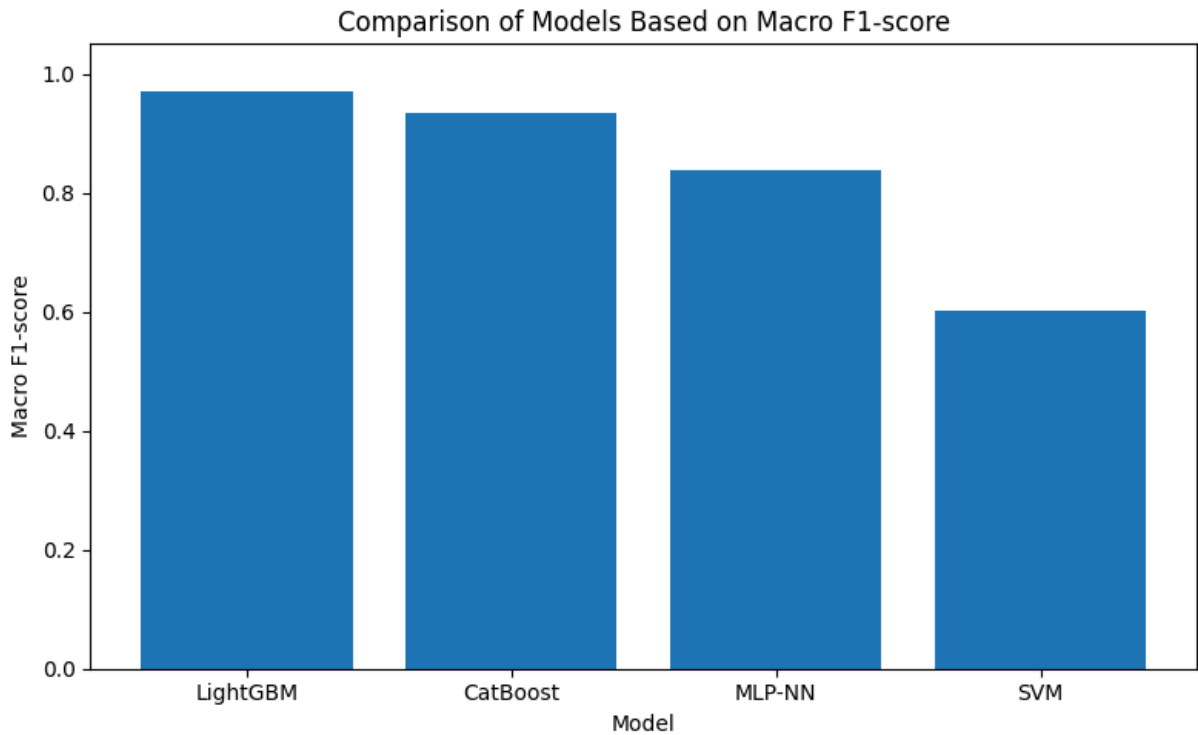


Fig. 2. Comparison of models based on Macro F1-score.

To further examine the best-performing model, the confusion matrix of the LightGBM Classifier is presented in Fig. 3. The model correctly classified most samples across all AQI classes. The largest number of correct predictions came from class 1 and class 2, which were also the most frequent classes in the dataset. However, LightGBM also showed strong performance on the less frequent higher-severity classes, correctly classifying most class 4 and class 5 observations. Most misclassifications occurred between neighboring AQI classes, such as class 3 and class 4 or class 4 and class 5, which is reasonable because main.aqi represents ordinal severity levels.

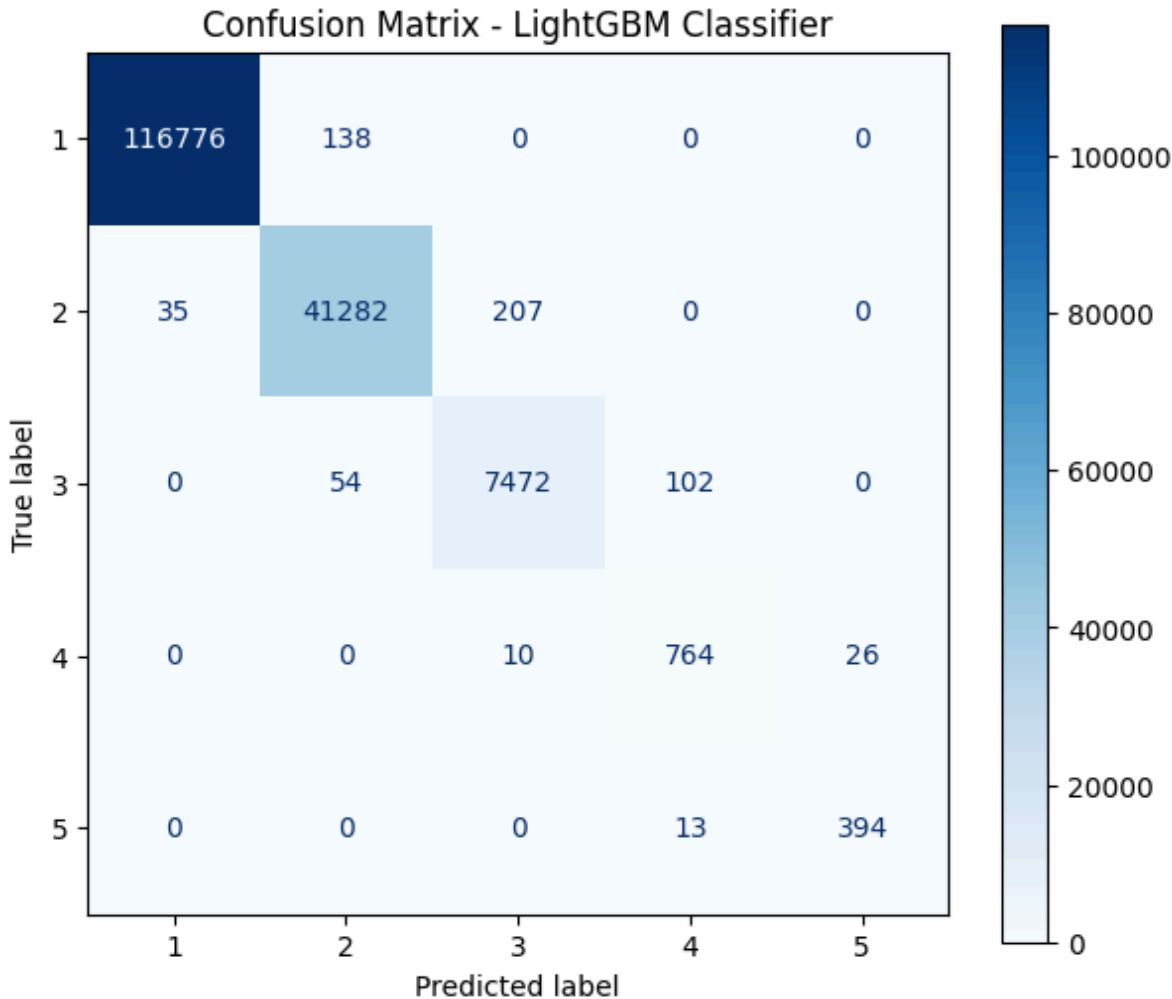


Fig. 3. Confusion matrix of the LightGBM Classifier on the test set.

The normalized confusion matrix in Fig. 4 provides a clearer view of class-level performance under class imbalance. LightGBM achieved approximately perfect classification for AQI classes 1 and 2, while maintaining strong normalized accuracy for classes 3, 4, and 5. Specifically, class 3, class 4, and class 5 achieved approximate normalized correct classification rates of 0.98, 0.96, and 0.97, respectively. This confirms that the model was not simply biased toward the dominant classes and was able to generalize well across minority AQI severity levels.

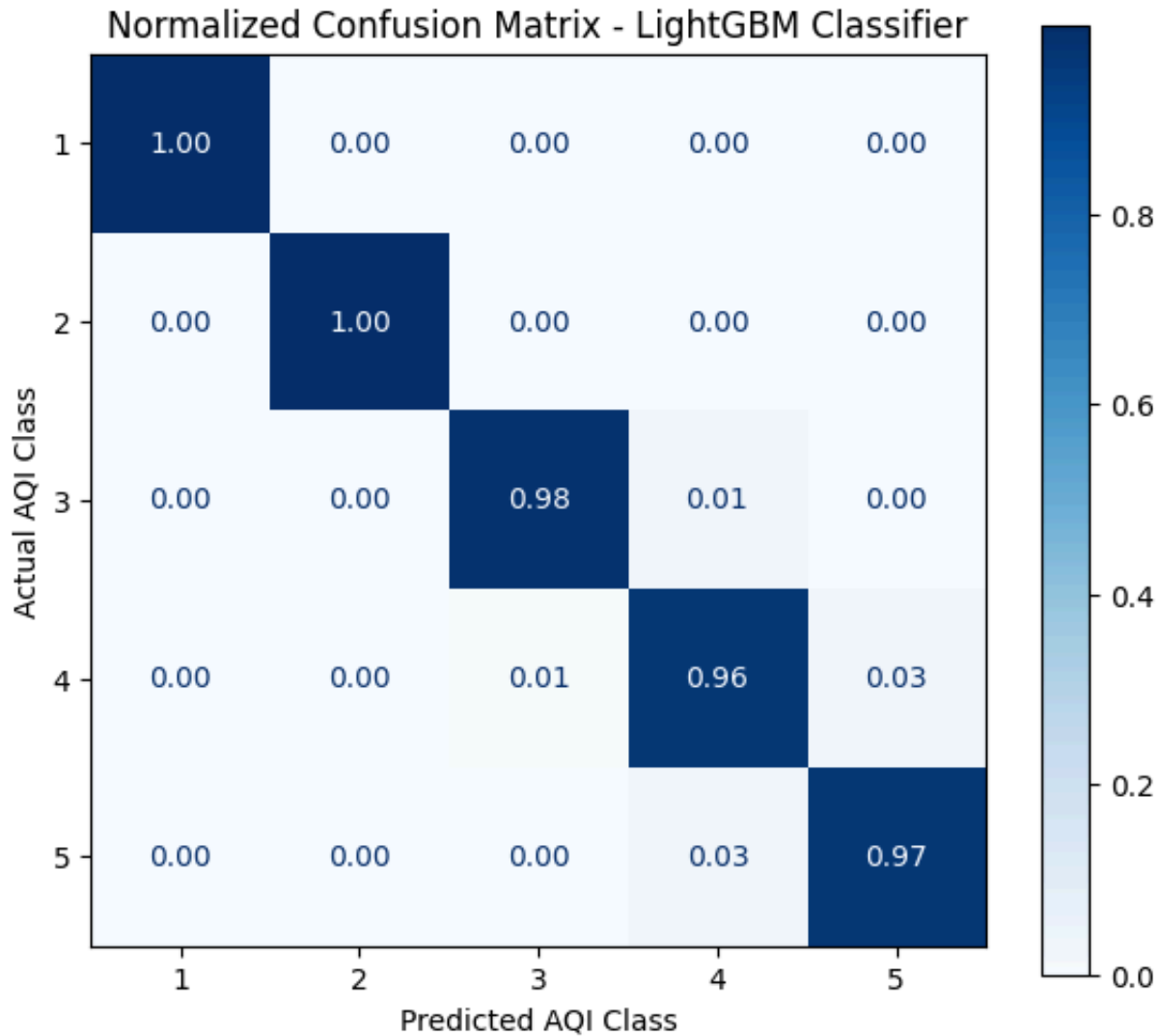


Fig. 4. Normalized confusion matrix of the LightGBM Classifier.

Finally, Fig. 5 shows the top 15 feature importances generated by the LightGBM Classifier. The highest-ranking features were primarily pollutant concentration variables, including NH_3 , NO_2 , CO , O_3 , SO_2 , NO , $\text{PM}_{2.5}$, and PM_{10} . This indicates that LightGBM relied mainly on air quality component values when distinguishing between AQI severity levels. Temporal variables such as hour, day of week, and month also contributed to the classification process, although their influence was lower than that of the pollutant features. City-based variables appeared with relatively small importance, suggesting that location provided additional context but was not the main basis of the model's predictions. These importance values should be interpreted as model-based indicators of predictive contribution rather than direct causal effects on AQI severity.

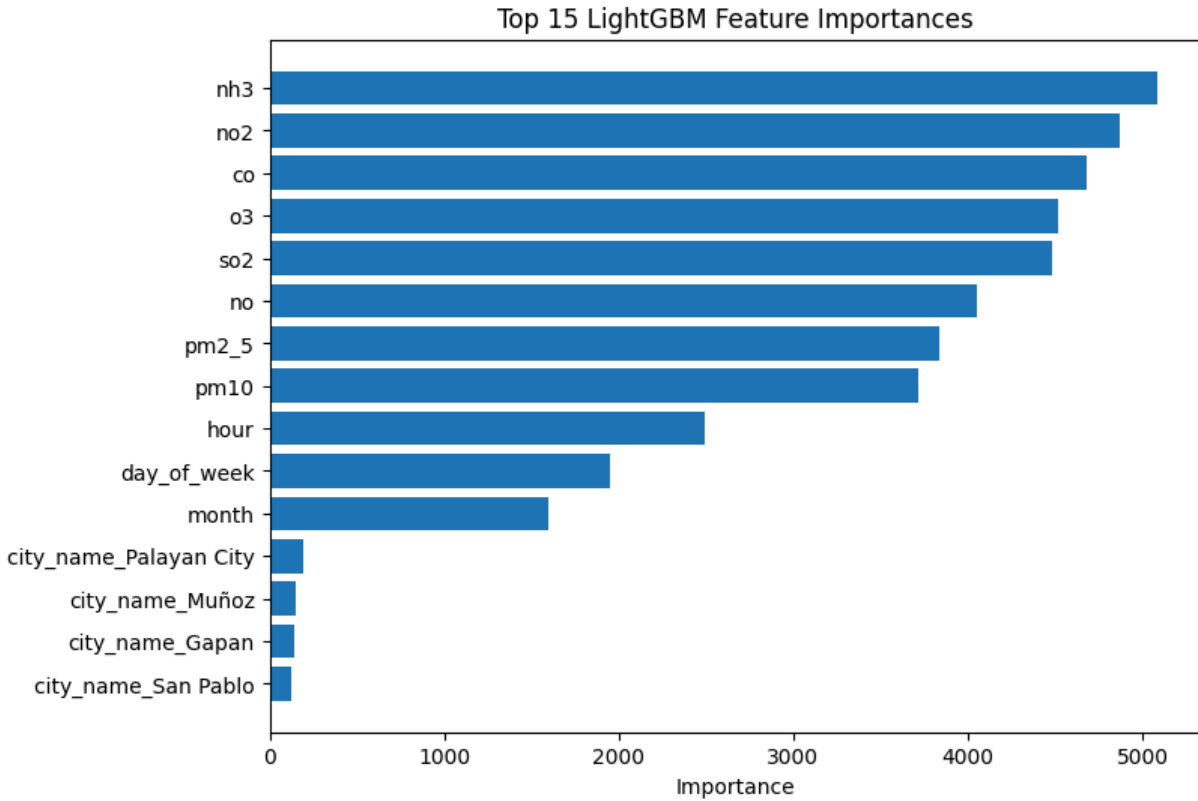


Fig. 5. Top 15 LightGBM feature importances.

Discussion

The results indicate that gradient boosting models were the most effective approaches for AQI level classification in this study. LightGBM achieved the highest overall performance, followed by CatBoost, MLP-NN, and SVM. This suggests that tree-based boosting models are well suited for structured air quality datasets because they can capture nonlinear relationships and interactions among pollutant concentration variables. Since AQI severity levels are influenced by multiple pollutants, models that can learn complex decision rules are more effective than simpler margin-based classifiers.

The strong performance of LightGBM is especially notable because the dataset is highly imbalanced, with most records belonging to lower AQI classes. Despite this imbalance, LightGBM achieved a high Macro F1-score, indicating that it performed well not only on the dominant classes but also on the less frequent AQI severity levels. This superior performance stems from LightGBM's leaf-wise tree growth strategy. By expanding the leaves with the highest loss, it isolates intricate patterns better than older level-wise models. This allowed LightGBM to capture subtle, non-linear differences between neighboring severity levels, such as AQI Class 4 and Class 5, which the SVM missed. The normalized confusion matrix further

supports this result, showing high correct classification rates across all AQI classes. Since `main.aqi` is an ordinal label, residual errors naturally occurred mostly between adjacent classes, which is less severe than misclassifications between distant severity levels.

CatBoost also performed strongly, which confirms the effectiveness of gradient boosting methods for AQI classification. However, LightGBM produced better macro precision, macro recall, and macro F1-score in this experiment. Although CatBoost is designed to handle categorical features effectively, the results suggest that the pollutant concentration variables were more influential than city-level categorical information. This is supported by the LightGBM feature importance results, where pollutant features ranked higher than city-based variables.

The MLP Neural Network achieved acceptable performance but did not outperform the gradient boosting models. This may be because the dataset is tabular in nature, where tree-based ensemble models often perform better than fully connected neural networks. Although the MLP-NN was able to learn nonlinear relationships, it likely required more careful tuning, architecture refinement, or feature representation to compete with the boosting models. The training and validation loss curves suggest that the neural network was able to train steadily, but its final classification performance remained lower than LightGBM and CatBoost.

The SVM baseline achieved the lowest performance among the four models. While SVM is useful as a classical classifier, its lower Macro F1-score suggests that it struggled with the nonlinear and imbalanced structure of the dataset. This supports the importance of comparing model families

rather than relying on a single algorithm. The difference between SVM and the boosting models shows that AQI level classification benefits from models capable of capturing feature interactions and class-specific decision boundaries.

The feature importance results show that pollutant concentration variables were the main contributors to AQI classification, while also suggesting possible proxy relationships among pollutants. Features such as NH_3 , NO_2 , CO , O_3 , SO_2 , NO , $\text{PM}_{2.5}$, and PM_{10} were the most important predictors in the LightGBM model. Although OpenWeather's Air Pollution API reports NH_3 and NO as pollutant components alongside CO , NO_2 , O_3 , SO_2 , $\text{PM}_{2.5}$, PM_{10} , and NH_3 , their high importance should not be interpreted as direct causal influence on AQI severity. Instead, the model may have used these variables as statistical indicators that help separate AQI classes within the dataset. For example, NH_3 and NO may reflect broader pollutant patterns or co-occurring emission conditions that are useful for classification, even if they are not always emphasized in standard AQI breakpoint tables. Similarly, the high importance of NO_2 , CO , O_3 , SO_2 , $\text{PM}_{2.5}$, and PM_{10} is consistent with the role of these pollutants in air quality assessment. Temporal variables such as hour, day of week, and month were less influential, suggesting that pollutant concentration patterns drove classification more strongly than time or city identity. Overall, these feature importance values should be treated as model-based indicators of predictive contribution rather than proof of direct environmental causation.

However, the study also has limitations. First, the dataset uses `main.aqi` as a five-level categorical AQI label rather than a continuous AQI value. Therefore, the study focuses on AQI severity classification

and does not predict exact AQI index values. Second, the target classes are imbalanced, with lower AQI classes appearing more frequently than higher-severity classes. Although Macro F1-score and confusion matrix analysis were used to address this issue, future work may explore additional class balancing strategies. Third, the dataset is limited to the available 2025 records from the selected Kaggle source, so the results should not be interpreted as a complete long-term assessment of Philippine air quality. Lastly, because AQI labels are derived from pollutant concentrations, the models may partly learn the AQI classification rules embedded in the dataset rather than discover independent environmental relationships.

Conclusion

This study demonstrates that machine learning can be used to classify AQI severity levels from Philippine city-level pollutant concentration data in a reproducible and comparative manner. Rather than treating AQI as a continuous forecasting problem, the study framed main.aqi as an ordinal multiclass target, which aligned with the structure of the selected dataset. This allowed the models to be evaluated based on their ability to distinguish between air quality severity categories using pollutant, temporal, and city-based features.

The findings highlight the value of model comparison in environmental data analysis. The results show that model choice significantly affects classification

performance, especially when the dataset contains imbalanced AQI classes. The stronger performance of gradient boosting methods suggests that structured air quality datasets benefit from models capable of learning nonlinear feature interactions. At the same time, the feature importance results indicate that pollutant concentration variables were more influential than temporal or city-based features, reinforcing the importance of pollutant-level data in AQI severity classification.

Although the study does not directly propose an intervention for reducing air pollution, it contributes to environmental health monitoring by providing an analytical pipeline for classifying AQI levels and interpreting model behavior. Such an approach may support future air quality studies that aim to compare models, examine pollutant contributions, or develop monitoring tools based on publicly available environmental data.

Future work may extend this study by using official monitoring station data, longer historical records, continuous AQI values, and additional meteorological variables such as temperature, humidity, wind speed, and rainfall. Further improvements may also include SHAP-based explainability, alternative class balancing methods, and time-based validation to evaluate how well the models generalize to future periods.

References

- Koons, E. (2024, June 13). *Air pollution in the Philippines – solutions*. Energy Tracker Asia.
<https://energytracker.asia/air-pollution-in-the-philippines/>
- Jcteamjoy. (2024, June 6). QC adds more air quality monitoring sensors; Boosts air quality management capability | Quezon City Government. *Quezon City Government*.
<https://quezoncity.gov.ph/qc-adds-more-air-quality-monitoring-sensors-boosts-air-quality-management-capability/>
- Air Quality Management | EMB-XI*. (n.d.).
<https://r11.emb.gov.ph/programs/air-quality-management/>
- Department of Science and Technology. (n.d.). Cordillera Administrative Region, “DOST, Partners Turn Over Project MINERVA,” DOST-CAR. [Online]. Available:
<https://car.dost.gov.ph/>
- Liu, H., Li, Q., Yu, D., & Gu, Y. (2019). Air Quality Index and Air Pollutant Concentration Prediction Based on Machine Learning Algorithms. *Applied Sciences*, 9(19), 4069.
<https://doi.org/10.3390/app9194069>
- Leong, W. C., Kelani, R. O., & Ahmad, Z. (2020). Prediction of air pollution index (API) using support vector machine (SVM). *Journal of Environmental Chemical Engineering*, 8(3), Article 103208.
<https://doi.org/10.1016/j.jece.2019.103208>
- Tirink, S. (2025). Machine learning-based forecasting of air quality index under long-term environmental patterns: A comparative approach with XGBoost, LightGBM, and SVM. *PLoS ONE*, 20(10), e0334252.
<https://doi.org/10.1371/journal.pone.0334252>
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In I. Guyon et al. (Eds.), *Advances in neural information processing systems* (Vol. 30). Curran Associates, Inc.
- Ravindiran, G., Hayder, G., Kanagarathinam, K., Alagumalai, A., & Sonne, C. (2023). Air quality prediction by machine learning models: A predictive study on the Indian coastal city of Visakhapatnam. *Chemosphere*, 338, Article 139518.
<https://doi.org/10.1016/j.chemosphere.2023.139518>
- Peinado, Luzneyda Ballesteros, et al. “Statistical and Machine Learning Models for Air Quality: A Systematic Review of Methods and Challenges.” *Algorithms*, vol. 18, no. 12, 11 Dec. 2025, pp. 783–783, www.mdpi.com/1999-4893/18/12/783,
<https://doi.org/10.3390/a18120783>.
- Gulati, Sumita, et al. “Estimating PM2.5 Utilizing Multiple Linear Regression and ANN Techniques.” *Scientific Reports*, vol. 13, no. 1, 19 Dec. 2023, p. 22578, www.nature.com/articles/s41598-023-49717-7,
<https://doi.org/10.1038/s41598-023-49717-7>.

Appendix

Appendix A. Group Member Contributions

Member	Role	Contribution
Eva, Kyle Francis E.	- Technical writer - Researcher - ML Developer	- Assisted in drafting the manuscript and peer reviewing. - Conducted foundational research, detailed methodology, and experimental results. - Created and structured the presentation in Canva. - Implemented the models in Colab and analyzed the results (AI-assisted).
Baron, Hans Schubert S.	- Peer Reviewer - Quality Assurance	- Conducted critical peer review of the initial manuscript to identify technical inconsistencies. - Assisted in refining the paper's scope and addressing dataset limitations. - Contributed

		to the writing of the Discussion section, providing deeper analysis of model behavior. - Ensured overall formatting and structural flow of the final document.
Duran, Vincent Gabriel D.	- Researcher - Writer - ML Assistant & Interpreter	- Assisted in drafting the manuscript particularly in the introduction and related literatures while also assisting in the discussion and conclusion - Assisted in interpreting the results and double checking the data gathered in Colab from the various models (AI-assisted)
Rodelas, Raymond J.	- Technical writer	- Assisted in drafting the manuscript, particularly the Related Works section.

Appendix B. Repository and Reproducibility Information

Item	Description
Repository	https://github.com/averageCoder-byte/ph-aqi-classification-ml.git
Notebook file	 CSELEC2C_Gro...
Dataset	PH Philippine Cities Air Quality Index Data 2025 from Kaggle
Main target variable	main.aqi
Main Task	Supervised multiclass AQI level classification
Main Models	SVM, LightGBM, CatBoost, MLP-NN
Presentation	 CSELEC2C_Gro...

		and MLP-NN
Implementation	ChatGPT	Assisted in drafting the implementation section, including dataset preparation, preprocessing, train-validation-test split, model training, hyperparameter tuning, evaluation metrics, and reproducibility details.

Appendix C. Application of LLM

Section	LLM	Contribution
Python notebook (Colab)	ChatGPT	Assisted on coding, analyzing, and interpreting
Model & Algorithm	ChatGPT	Assisted in explaining the assumptions and mechanisms of SVM, LightGBM, CatBoost,